

Agentic AI-Based Smart Decision Support System for Electricity Procurement and Market Trading for Kerala State Electricity Board



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PROJECT GUIDE

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MOTIVATION



- Electricity demand varies due to weather, seasons, and consumer behavior.
- Large volumes of operational data make manual analysis difficult.
- Existing systems provide monitoring and forecasting but limited decision support.
- Many decisions still depend on operator experience and static thumb rules rather than optimized data-driven analysis.
- An AI-powered system can generate insights, recommendations, risk alerts, and adaptive thumb rules for better decision-making.



OBJECTIVES

Demand & Price Forecasting : Forecast short-term electricity demand and market prices.

Hydro Resource Assessment: Estimate future reservoir inflows and available hydro resources.

MILP-Based Optimal Scheduling : Generate cost-effective hydro and pumped-storage schedules.

RL-Based Storage Control: Learn intelligent pump/generate strategies for Pumped Storage Plant operation.

Explainable AI Decision Support: Provide validated recommendations and operator-friendly explanations.

LITERATURE SURVEY

Paper	Details	Method	Key Bottleneck Addressed	Research Gap
Short-Term Load Forecasting: Review + CNN-LSTM Hybrids	K. Ullah et al., IEEE Access (2024)	Hybrid CNN-LSTM (CNN spatial features + LSTM temporal; MAPE $\approx$ 2.7%)	Accurate short-term demand forecasting under volatile weather/seasonal load	Load-only; no price forecasting, scheduling, or decision support
MFformer: Transformer-Based Multi-Frequency Feature Aggregation for Load Forecasting	H. Tong, J. Liu, Electric Power Systems Research, vol. 243 (2025)	Transformer with multi-frequency feature aggregation; self-attention captures multi-scale temporal patterns	Long-range, multi-frequency demand patterns that CNN-LSTM can miss	Pure forecasting; no link to MILP/RL optimization or market decisions
Deep Learning-Based Electricity Price Forecast for Virtual Bidding	X. Wang et al., IEEE Transactions on Power Systems, vol 37 (2024)	Transformer encoder forecasting DA–RT price spread (ERCOT) from load/solar/wind forecasts; walk-forward + backtested trading strategies	Market price forecasting under renewable volatility + linking forecast to trading profit	Price + simple trading rules only; no hydro/storage scheduling, RL control, or DT validation
Optimization of Sizing and Operation of Pumped Hydro Storage Plants	G. A. M. Castorino et al., Journal of Energy Storage (2025)	Surrogate-Modeling Optimization (SMO) + MILP solved simultaneously over current/future price scenarios	Cost-optimal pumped-storage sizing & operation under economic uncertainty	Planning/economic optimization; no real-time RL control, forecasting front-end, or decision support
Distributionally Robust Day-Ahead Scheduling of Cascade Pumped Hydro Storage	M. Liu et al., Applied Energy, vol. 384 (2025)	Two-stage DRO (moment-based ambiguity set) reformulated to MILP; cascade PHES + DC transmission	Robust day-ahead hydro/PHES scheduling under wind/solar forecast uncertainty	Optimization only - no learning/adaptation, RL, DT, or operator-facing decision support
DRL-Based Medium-Term Planning of Cascaded Hydropower for Wholesale Market	X. Chen, Y. Liu, N. Fan, L. Wu, Water (2025)	DRL + multi-parametric programming for self-scheduling cascaded hydropower guiding market participation	Replacing static "rules of thumb" with data-driven hydro planning that maximizes market profit	Medium-term planning only; no short-term pump/generate control, DT validation, or explainable interface
Optimizing Bidding Strategy via GCN + Deep RL	H. Weng et al, Applied Energy, vol. 380 (2025)	Graph Convolutional Network + DDPG (multi-agent) capturing inter-participant correlations	Learning strategic bidding / market trading in continuous action space under uncertainty	Bidding only; no physical hydro/PHES constraints, forecasting, DT, or human-in-the-loop

Paper	Details	Method	Key Bottleneck Addressed	Research Gap
RL Real-Time Control of a Reversible Pump Turbine	C. Tubeuf, et al., <i>Energy and AI</i> , vol. 24 (2026)	PPO agent trained on a Digital Twin (sim-to-real) for pump start-up and rule-based surrogate for safety/interpretability (>30% faster start-up)	Real-time pump/generate control of pumped storage + DT-based safe validation + interpretability	Unit-level control only; not coupled to demand/price forecasts, MILP schedules, or market/decision layer
GridMind: LLMs-Powered Agents for Power System Operations	H. Jin, et al. SC'25 Workshops (2025)	Multi-agent LLM orchestrating deterministic solvers via function calls; conversational what-if decision support	Agentic AI augmenting operator decisions and integrating tools via natural language	Steady-state analysis only; no forecasting / MILP-hydro / RL-storage integration, no market trading, no DT
Grid-Agent: LLM-Powered Multi-Agent System for Grid Control	Y. Zhang et al., SC(2025)	LLM multi-agent (planner + validation agent) + power-flow solvers + MARL; sandboxed validation with rollback + human-readable explanations	Explainable, validated, autonomous control with human-in-the-loop safety	Grid-violation focus; no hydro/PHES scheduling, demand/price forecasting, or market-trading orientation
Redesigning Grid Dispatching by Integrating LLMs, KGs & AI Agents	G. Liu, Y. et al. CSEE JPES, vol. 11(6) (2025)	LLM + Knowledge Graph + AI agents; closed-loop autonomous dispatching for data interoperability, decision transparency, adaptability	Transparent, adaptive agentic dispatching decision support	Conceptual dispatching automation; not validated for hydro market trading; no MILP/RL/DT/forecast specifics
Towards an Explainable AI Approach for Smart Grid Systems	M. Alfayan et al. Discover AI, vol. 5 (2025)	Type-2 fuzzy-logic XAI generating interpretable energy-demand predictions	Operator-friendly explainable predictions/ decision support	Demand-prediction explainability only; no scheduling/optimization/control or unified pipeline
Short-Term Hydropower Regulation via Coupled Hydrological–Energy Models: A Systematic Review	S. M. Vatanchi, et al., R&SE Reviews, vol. 235 (2026)	PRISMA review (106 studies) of hydro–energy–market coupling; taxonomy of coupling structure, uncertainty, solution methods	Defines the integration gap across forecasting, scheduling, market & validation	Their finding = our opportunity: coupling is one-way, integrated water–power feedback is rare, operational validation is limited; calls for ML-forecasting + DT-enabled validation + decision support
Fault Detection & System Resilience in Hydropower using Digital Twins + Deep Learning	J. Tan et.al, Scientific Reports, vol. 15 (2025)	Digital Twin + Deep Learning hybrid; real-time modeling + fault prediction (12.14% faster fault detection)	DT-based simulation/validation and risk alerting before execution	Fault detection/monitoring only; no schedule optimization, market trading, or RL/MILP/forecast integration
Deep RL for Optimized Reservoir Operation under Uncertainty	F. Sseguya, K. J. Jun, Water, vol. 17(22) (2025)	DRL (DQN/PPO/DDPG) integrating precipitation/temperature/soil-moisture for adaptive storage decisions (Soyang Dam)	Reservoir/hydro resource assessment + adaptive storage control under inflow uncertainty	Single-reservoir, water-supply focus; no electricity price/market coupling, MILP, DT, or LLM decision support

Existing Research

- Forecasting (Refs 1–3), MILP/robust optimization scheduling (Refs 4–6), RL control & market bidding (Refs 6–8, 15), Digital Twins (Refs 8, 14), and agentic / explainable decision support (Refs 9–12) are each well-developed in isolation.
- The 2026 systematic review (Ref 13) confirms that coupling between hydrology, scheduling, market, and validation remains predominantly one-way, with iterative/fully-integrated water–power feedback rare and rigorous operational validation limited.

Research Gap

- No published framework unifies Forecasting + MILP scheduling + RL pumped-storage control + Digital-Twin validation + LLM/agentic explainable decision support with human-in-the-loop, tailored to a hydro-dominant utility's procurement and market trading (KSEB).

Specific unaddressed needs we target:

1. An end-to-end forecast → optimize → control → validate → explain pipeline (no prior work closes this loop).
2. Digital-Twin pre-execution validation of schedules via historical replay and scenario/regret analysis.
3. Operator-centric explainability and adaptive thumb rules that replace static, experience-based rules and reduce manual analysis.
4. Coupling RL real-time pump/generate control to MILP day-ahead schedules and price/demand forecasts within one agentic system.

RESEARCH GAP



SYSTEM OVERVIEW

FORECASTING & OPTIMIZATION

Forecasting Module

- Collects historical demand, market price, weather, and reservoir data.
- Predicts short-term electricity demand using time-series forecasting models.
- Forecasts future electricity market prices for scheduling decisions.
- Estimates reservoir inflows and available hydro resources

Optimization Module

- Generates cost-optimal day-ahead scheduling plans using MILP.
- Considers reservoir constraints, generation limits, and DSM (Deviation Settlement Mechanism) penalties.
- Determines optimal generation, pumping, and procurement schedules.

Output of Forecasting & Optimization

- Demand Forecast
- Price Forecast
- Reservoir Forecast
- Optimal Dispatch Schedule



RL-Based Pumped Storage Control

- Learns optimal pump/generate decisions under varying demand and price conditions.
- Improves pumped-storage utilization and operational efficiency.

Digital Twin & Validation

- Simulates real-world operating conditions.
- Evaluates schedules through historical replay and scenario analysis.
- Quantifies cost savings and operational improvements.

AI Decision Support Layer

- Generates ranked operational recommendations.
- Provides explainable insights for every suggested action.
- Identifies risks and generates operator-friendly thumb rules.

Human-in-the-Loop

- Operators review and approve all recommendations before execution.

Final Output

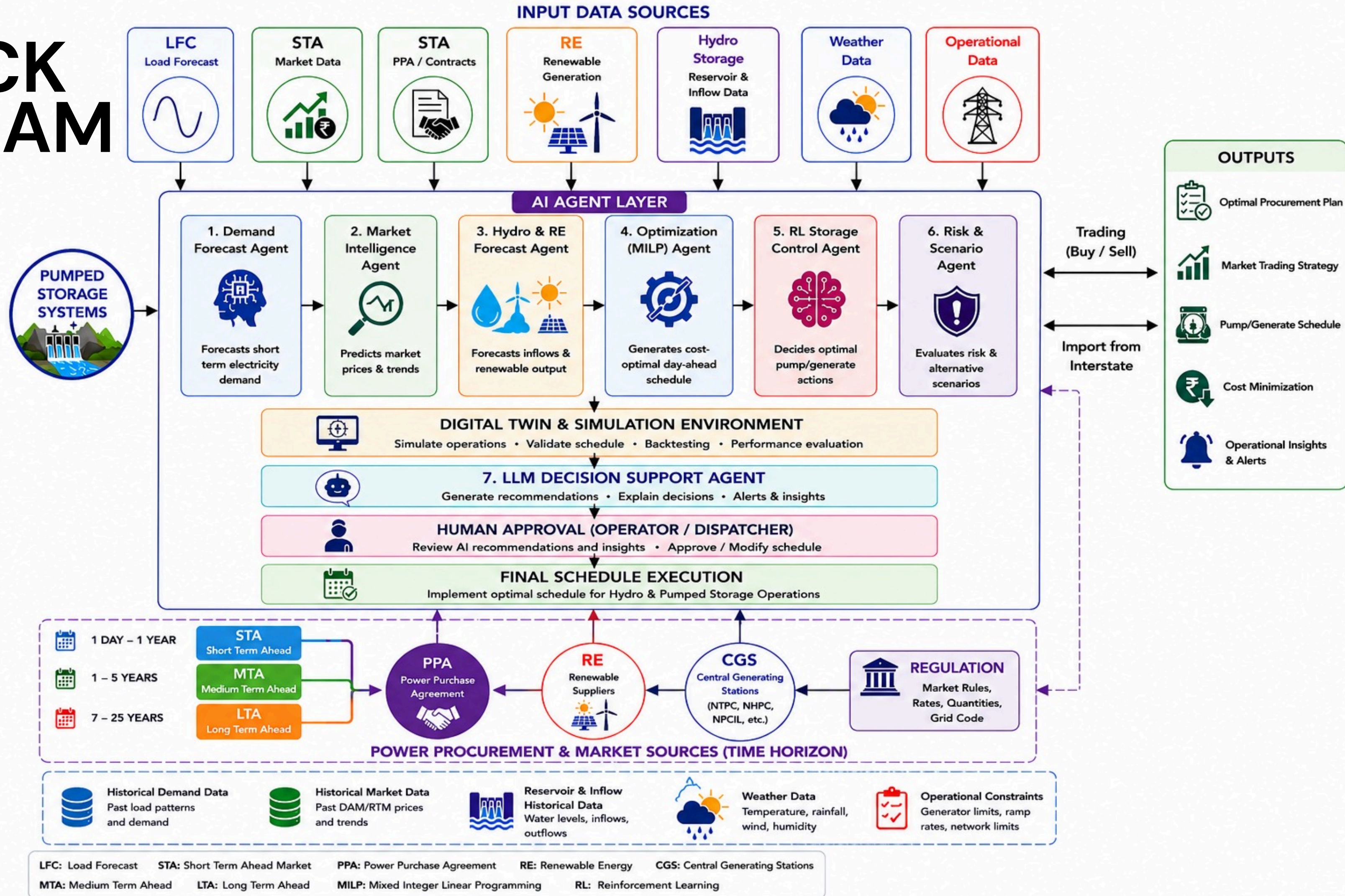
- Optimized Schedules
- Risk Alerts
- Explainable Recommendations
- Decision Support Dashboard

SYSTEM OVERVIEW

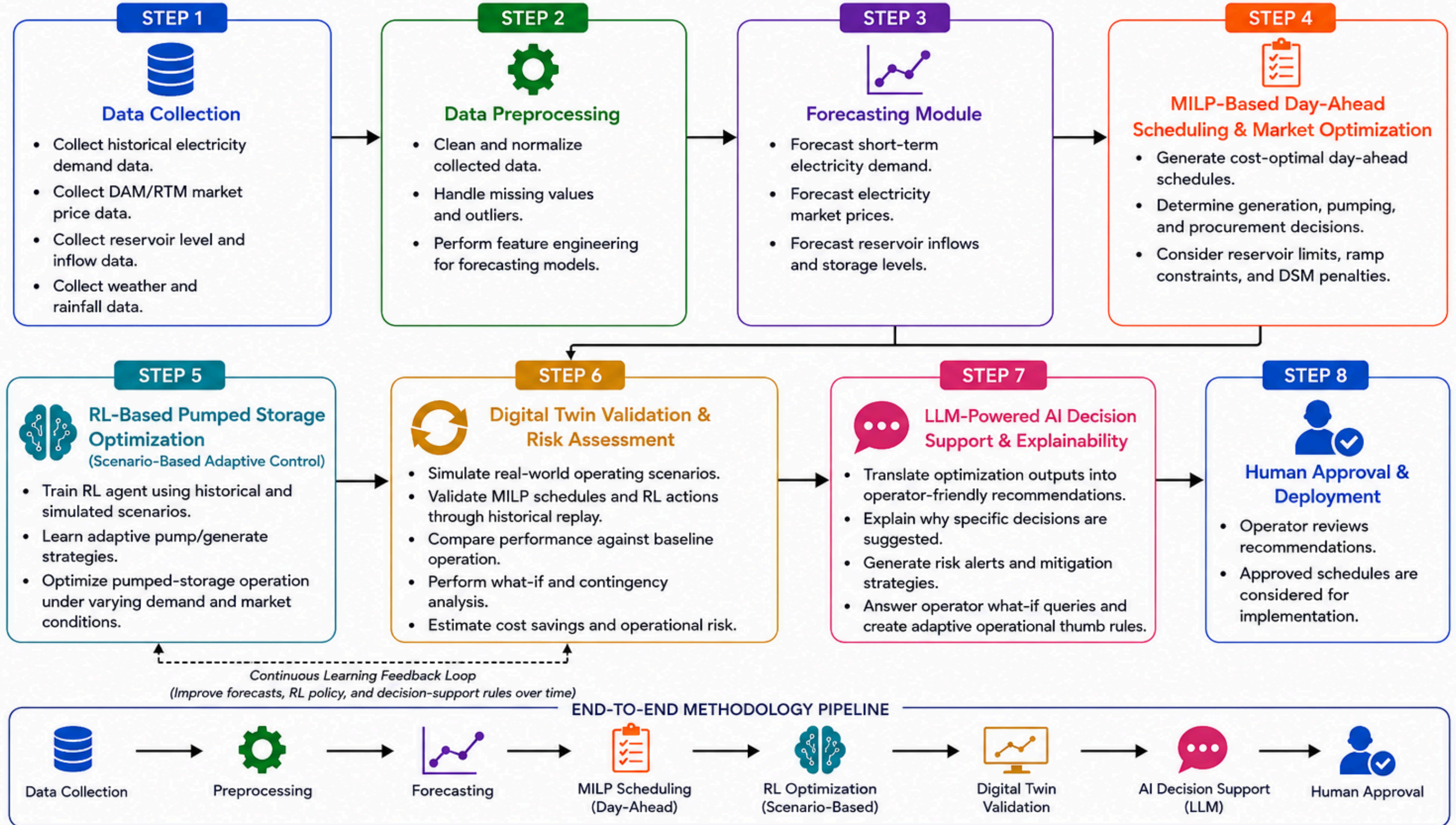
AI DECISION SUPPORT



BLOCK DIAGRAM



METHODOLOGY



This integrated framework combines Forecasting, Optimization, Reinforcement Learning, Digital Twin Validation, and Explainable AI to provide cost-effective, reliable, risk-aware, and transparent decision support for KSEB operations (using historical and simulated data in Phase 1).

EXPECTED OUTCOMES

- **Intelligent Decision-Support System for KSEB** that predicts future demand, market prices, and reservoir conditions, generates optimal schedules, validates decisions through simulation, and provides explainable recommendations before execution.
- **Accurate Forecasting Framework** for electricity demand, market prices, and reservoir inflows to support proactive planning and resource management.
- **Cost-Optimized Day-Ahead Scheduling** using MILP for hydro generation, pumped-storage operation, and electricity procurement decisions.
- **Scenario-Based Adaptive Pumped Storage Optimization** using Reinforcement Learning trained on historical and simulated operating conditions.
- **Digital Twin-Based Validation Environment** for scenario testing, risk assessment, performance evaluation, and safe validation of scheduling decisions.
- **Explainable AI & Human-in-the-Loop Decision Making** providing transparent recommendations, risk alerts, operational insights, and operator approval before implementation.
- **Unified Agentic AI Framework** integrating Forecasting, Optimization, Reinforcement Learning, Digital Twin Validation, and Explainable AI into a single workflow for smarter and more reliable grid operations.

TECH STACK

Backend & Frontend



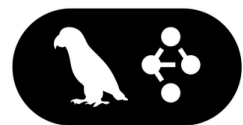
ML & Forecasting



Digital Twin



AI Agent Layer & Explainability



LangGraph



Gemini

COST ESTIMATE OF THE PROJECT

Project Cost Breakdown

- AI API Costs & Development – ₹3,000
- Testing & Deployment – ₹3,500
- Travel Expense for Site Visits – ₹2,500

Grand Total – ₹9,000

TIME FRAME FOR EXECUTION

(JULY 2026 – APRIL 2027)

PERIOD	ACTIVITIES	EXPECTED OUTCOME
 July 2026	Dataset collection, preprocessing, cleaning, and exploratory analysis	 Cleaned and validated datasets ready for development
 August 2026	Feature engineering and forecasting model design	 Forecasting framework and input features finalized
 September 2026	Demand forecasting model development and evaluation	 Accurate electricity demand forecasting model
 October 2026	Market price and reservoir inflow forecasting development	 Market and hydro forecasting modules completed
 November 2026	MILP-based hydro scheduling model development	 Cost-optimal scheduling framework developed
 December 2026	Enhancement of MILP model with operational constraints and testing	 Validated scheduling model for hydro operations
 January 2027	RL-based pumped storage control development	 Intelligent pump/generate control policy learned
 February 2027	Digital Twin simulation and backtesting framework development	 Simulation environment for schedule validation completed
 March 2027	LLM-based decision support, explainability, and recommendation modules	 AI-generated recommendations, insights, and risk alerts
 April 2027	System integration, testing, documentation, and final evaluation	 Fully integrated AI-driven decision support system ready for deployment and presentation

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THANK YOU